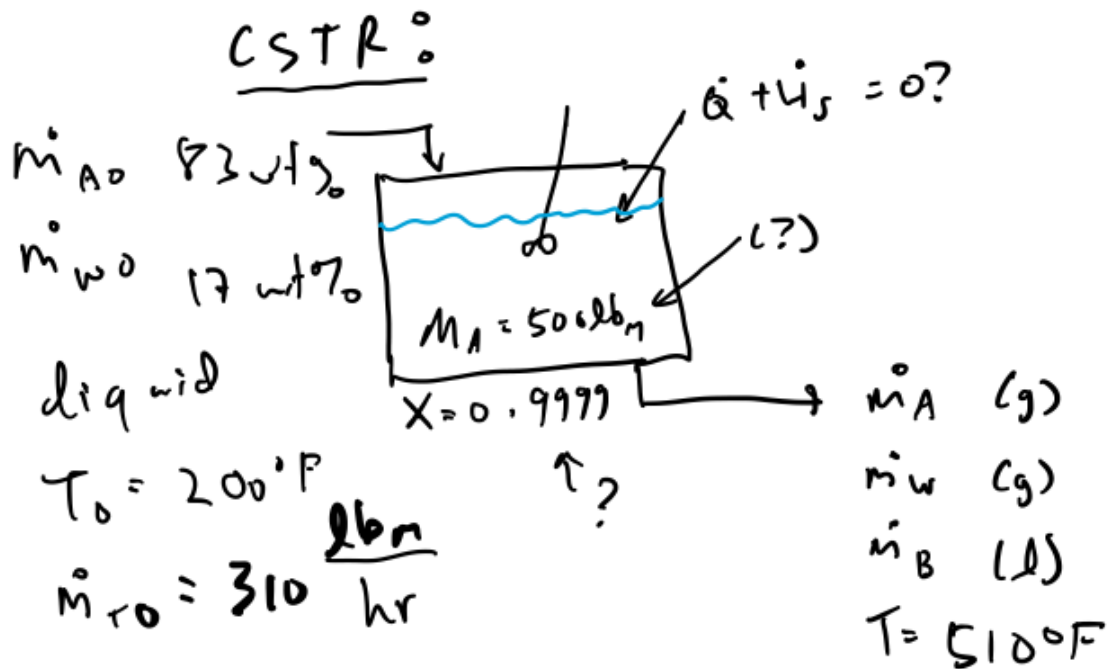
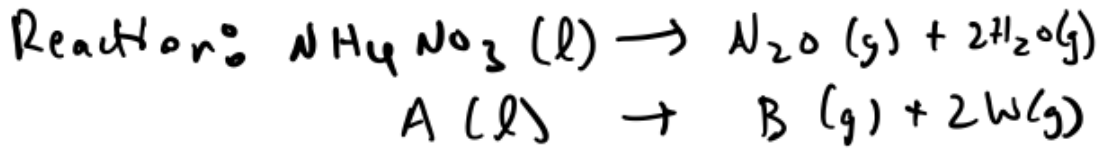


21 Nitrous Oxide Plant Example (content warning: #explosion)

Tuesday, October 22, 2024
9:13 PM



$-r_A = kM_A$, units = $\frac{\text{lb}_m}{\text{min}}$

mass balance:

$0 = \dot{m}_{A0} - \dot{m}_A - kM_A$

$\therefore M_A = \frac{\dot{m}_{A0} - \dot{m}_A}{k}$

$k = 0.00517 \text{ min}^{-1} @ 510^\circ\text{F} = 0.307 \text{ hr}^{-1}$

$$\dot{m}_{A0} = \left(310 \frac{\text{lbm}}{\text{hr}}\right) (0.83)$$

$$\dot{m}_A = \dot{m}_{A0} (1 - X) = \left(310 \frac{\text{lbm}}{\text{hr}}\right) (0.83) (1 - 0.9999)$$

$$\therefore M_A = \frac{\left(310 \frac{\text{lbm}}{\text{hr}}\right) (0.83) (1 - (1 - 0.9999))}{0.307 \text{ hr}^{-1}}$$

$M_A = 829 \text{ lbm A}$ in the reactor

Analyze ΔH_{rxn}

$$\Delta H_{rxn} = -336 \frac{\text{Btu}}{\text{lbm}} @ 500^\circ\text{F} \quad \text{constant?}$$

$$C_{pA} = 0.38 \frac{\text{Btu}}{\text{lbm} \cdot ^\circ\text{F}} = 30.4 \frac{\text{Btu}}{\text{lbmol} \cdot ^\circ\text{F}}$$

$$C_{pW} = 0.47 \frac{\text{Btu}}{\text{lbm} \cdot ^\circ\text{F}} = 8.8 \frac{\text{Btu}}{\text{lbmol} \cdot ^\circ\text{F}}$$

$$C_{pB} = ? \quad B = \text{N}_2\text{O} \quad \text{Assume it's}$$

an ideal gas. For a polyatomic ideal

$$\text{gas, } C_p = 4R = 4 \left(1.986 \frac{\text{Btu}}{\text{lbmol} \cdot ^\circ\text{F}} \right) = 7.9 \frac{\text{Btu}}{\text{lbmol} \cdot ^\circ\text{F}} = 0.18 \frac{\text{Btu}}{\text{lbm} \cdot ^\circ\text{F}}$$

$$\Delta C_p = C_{pB} + 2 C_{pH} - C_{pA} = -5.5 \frac{\text{Btu}}{\text{lbmol} \cdot ^\circ\text{F}} = -0.07 \frac{\text{Btu}}{\text{lbm} \cdot ^\circ\text{F}} \leftarrow \text{very small compared to } \Delta H_{\text{rxn}}, \text{ so assume } \Delta H_{\text{rxn}} \text{ is constant w/ } T$$

energy balances

$$0 = \dot{m}_{A0} H_{A0} + \dot{m}_{W0} H_{W0} - \dot{m}_A H_A - \dot{m}_H H_H - \dot{m}_B H_B + (\dot{Q} + \dot{W}_s)$$

$$\dot{m}_H = \dot{m}_{W0} + \frac{2 \dot{m}_{A0}}{MW_A} MW_W = \dot{m}_{A0} \left(\Theta_H + 2 \frac{MW_W}{MW_A} X \right), \quad \Theta_H = \frac{\dot{m}_{W0}}{\dot{m}_{A0}} = \frac{0.17}{0.83}$$

$$0 = \dot{m}_{A0} H_{A0} + \dot{m}_{W0} H_{W0} - \dot{m}_{A0} (1-X) H_A$$

$$- \dot{m}_{A0} \left(\Theta_H + 2 \frac{MW_W}{MW_A} X \right) H_H - \dot{m}_{A0} H_{N20} \left(\frac{MW_{N20}}{MW_A} \right) X$$

$$+ (\dot{Q} + \dot{W}_s)$$

$$0 = \dot{m}_{A0} (H_{A0} - H_A) + \dot{m}_{W0} (H_{W0} - H_H)$$

$$+ \dot{m}_{A0} X \left(H_A - 2 \frac{MW_W}{MW_A} H_H - \frac{MW_{N20}}{MW_A} H_{N20} \right) + (\dot{Q} + \dot{W}_s)$$

$$0 = \dot{m}_{A0} C_{pA} (T_0 - T) + \dot{m}_{W0} (H_{W0} - H_H) +$$

$$\dot{m}_{A_0} X (-\Delta H_{rxn}) + (\dot{Q} + \dot{W}_s)$$

get $H_{w_0} - H_{w_0}^\circ$

feed:

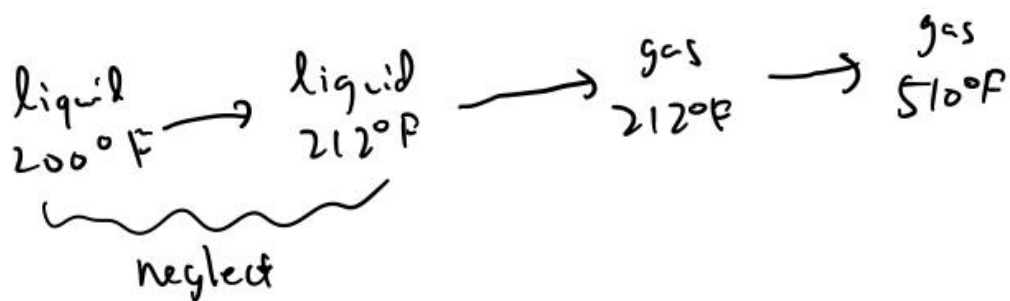
$$T = 200^\circ\text{F}$$

liquid

exit:

$$T = 510^\circ\text{F}$$

gas



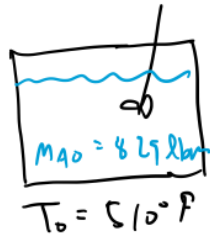
$$H_{w_0} - H_{w_0}^\circ = H_{f,w(l)}^\circ - H_{f,w(g)}^\circ + C_{pw} (212^\circ\text{F} - 510^\circ\text{F})$$

$$\begin{aligned} 0 = & \dot{m}_{A_0} C_{pA} (T_0 - T) + \dot{m}_{w_0} (H_{f,w(l)}^\circ - H_{f,w(g)}^\circ) \\ & + \dot{m}_{w_0} C_{pw} (212^\circ\text{F} - 510^\circ\text{F}) + \dot{m}_{A_0} X (-\Delta H_{rxn}) \\ & + (\dot{Q} + \dot{W}_s) \end{aligned}$$

$$\therefore \dot{Q} + \dot{W}_s = 5739 \frac{\text{Btu}}{\text{hr}} = 2.25 \text{ hp}$$

feed shot off

model like an adiabatic batch reactor:



mass balance: $M_{A0} \frac{dX}{dt} = k M_A = k M_{A0} (1-X)$

$$\therefore \frac{dX}{dt} = k(1-X)$$

$$k = (0.00517 \text{ min}^{-1}) \exp \left(44520^\circ \text{R} \left(\frac{1}{510 + 460^\circ \text{R}} - \frac{1}{T + 460^\circ \text{R}} \right) \right)$$

energy balance:

$$T = \frac{T_0 \sum \theta_i C_{pi} + X (-\Delta H_{rxn} + \Delta C_p \cdot 500^\circ \text{F})}{\sum \theta_i C_{pi} + X \Delta C_p}$$

reference T for ΔH_{rxn}

$$\sum \theta_i C_{pi} = 0.38 \frac{\text{Btu}}{\text{lb}_m \text{A}^\circ \text{F}} + \frac{0.17 \text{ lb}_m \text{W}}{0.193 \text{ lb}_m \text{A}} \left(0.47 \frac{\text{Btu}}{\text{lb}_m \text{W}^\circ \text{F}} \right)$$

$$= 0.48 \frac{\text{Btu}}{\text{lb}_m \text{A}^\circ \text{F}}$$